

As a manuscript



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TECHNOLOGY AND EQUIPMENT FOR THE SYNTHESIS OF
COLLOIDAL QUANTUM DOTS
CsPbX₃ (X = Cl, Br, I) AND CdSe/ZnS,
Ag/SiO₂ PLASMONIC NANOPARTICLES,
AND HYBRID STRUCTURES BASED ON THEM

Specialty 05.27.06 — Technology and Equipment
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and Electronic Devices

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The Russian text is authoritative:

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The thesis is available in the library of Saint Petersburg Electrotechnical University “LETI” named after V. I. Ulyanov (Lenin) and on the university website www.eltech.ru (section “Science” — “Training of Highly Qualified Personnel” — “Announcements of Doctoral and Candidate Thesis Defenses”).

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Scientific Secretary of
Dissertation Council D 212.238.04,
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General Description of the Work

Relevance of the research. Structures with linear dimensions of 1–100 nm substantially change how electromagnetic radiation in the optical range propagates. For example, the localized surface plasmon resonance (LSPR) of metal nanoparticles produces a local enhancement of the electromagnetic field near their surface. A quantum emitter can be placed in this field: an organic fluorophore molecule or a quantum dot (QD). In such a hybrid system, the photoluminescence (PL) of a QD can be either enhanced in the local field of a plasmonic particle or quenched when the two are in direct contact. The distance at which plasmonic enhancement prevails over quenching is set by the thickness of the dielectric layer (spacer) between the plasmonic and the quantum subsystems. Plasmonic enhancement manifests itself differently in different hybrid structures [1, 2], and it remains relatively poorly studied.

Semiconductor and metal nanoparticles have many applications, both in combination and separately. In biomedicine, such particles are of interest as a basis for biomarkers with high contrast and chemical stability. However, the technology of synthesizing nanostructures with a specified set of functional properties still faces open problems: spectral range, quantum efficiency, biocompatibility, radiation hardness, and others. Active research directions also include the search for methods to create new materials and the use of the energy transfer effects described above between nano-objects of different composition to improve the output characteristics of hybrid nanostructures. Such effects are important for the amplification of stimulated emission in solid-state lasers, for higher solar cell efficiency, for active elements of plasmonics and photonics, and for the corresponding structures of flexible electronics.

Therefore, the topic of this thesis—the synthesis and study of colloidal quantum dots, plasmonic nanoparticles, and hybrid structures based on them—is relevant and of scientific and practical interest.

The aim of this work was to develop the physical and technological principles of the colloidal synthesis of low-dimensional semiconductor and metal nanostructures in polar and nonpolar dispersion media, and to fabricate hybrid structures based on them and study their optical properties.

To achieve this aim, the following **objectives** were addressed:

1. To study the formation of semiconductor and metal nanoparticles of different chemical composition during colloidal synthesis.
2. To study the formation of solid layered structures based on colloidal nanoparticles.
3. To study the plasmonic enhancement of photoluminescence in hybrid structures based on quantum dots and plasmonic nanoparticles.

The practical significance of the thesis is as follows:

1. A hardware system was developed. It includes a) a flow reactor for the synthesis of semiconductor and metal nanoparticles, b) an experimental setup for real-time measurement of light-scattering indicatrices, c) an experimental setup for successive ionic layer adsorption and reaction (SILAR) and electrochemical atomic layer deposition (E-ALD) under a variable external electric field. These engineering solutions are protected by three Russian utility model patents.

2. The photoluminescence of CsPbBr₃ perovskite nanocrystals was found to be radiation-hard at high absorbed doses of γ -radiation. This makes CsPbBr₃ of practical interest as a photoluminescent material for high-radiation environments.

3. Halide perovskite CsPbX₃ (X = Cl, Br, I) nanocrystals were shown to order efficiently when deposited from colloidal solutions. This is of interest for the active layers of light-emitting structures.

4. The photoluminescence intensity of A^{II}B^{VI} and A^{IV}B^{VI} semiconductor nanocrystals was shown to increase substantially when they are in contact with monodisperse Ag/SiO₂ nanoparticles and the photoluminescence of the hybrid system is excited at the LSPR frequency.

Main statements for defense:

1. The triangulation of the Cs–Pb–X system (X = Cl, Br, I) and the growth of CsPbX₃ nanocrystals are governed not only by the binary compounds on the edges of the Gibbs triangle, but also by the composition points that correspond to the [PbX₆] anionic octahedra.

2. The physicochemical features of the Cs–[PbX₆]–[PbY₆] interaction (X, Y = Cl, Br or Br, I) predetermine the efficient formation of cesium lead halide solid-solution nanocrystals and the change of their photoluminescence spectra during anion exchange.

3. A model of photoluminescence in cesium lead halide perovskites, in which the photoluminescence is determined by the anionic octahedra as structural units, explains their radiation hardness. This effect gives CsPbBr₃ nanocrystals an advantage over CdSe/ZnS quantum dots in high-radiation environments.

4. The photoluminescence enhancement in a hybrid system of CdSe/ZnS core/shell quantum dots and Ag/SiO₂ plasmonic nanoparticles arises from a manifold increase in the extinction of the combined system. The effect is controlled by varying the thickness of the dielectric shell.

Scientific novelty. All the main statements for defense are new.

The reliability of the results is supported by the reproducibility of the experimental data on different materials, and by comparison of the results with previously known theoretical models and published data.

Implementation of the results. The results were implemented in the Confocal Microscopy Laboratory of the Department of Pathomorphology of the D. O. Ott Research Institute of Obstetrics, Gynecology and Reproduc-

tology, within the Russian Science Foundation grant 14-15-00324 “Colloidal quantum dots as biomarkers in exploratory research on pathological processes of the female reproductive system” (2014–2016); in the Laboratory of Nonequilibrium Processes in Semiconductors of the Ioffe Institute, within joint research in 2017–2018; and in the teaching at Saint Petersburg Electrotechnical University “LETI”, in the courses “Materials Science of Micro- and Nanosystems” and “Nanomaterials” of the master’s program “Nanotechnology and Diagnostics” (field of study 28.04.01 “Nanotechnologies and Microsystem Engineering”).

The results of the thesis were used in the following projects: agreement 14.132.21.1703 of the Federal Target Program “Research and Scientific-Pedagogical Personnel of Innovative Russia (2009–2013)” on “Computer modeling of materials based on carbon nanotubes and lead sulfide quantum dots”; the UMNİK program project 3578GU2/2014 “Synthesis, study, and modeling of the formation of colloidal lead sulfide quantum dots”; the project part of the state assignment of the Ministry of Education and Science of the Russian Federation No. 16.2112.2014/K “Fabrication and study of porous systems functionalized with nanomaterials for photonics, sensing, and medicine”; and the Russian Foundation for Basic Research grant 17-33-80010 “Metal coatings on porous materials by additive technologies”.

Objects and methods of research. Unless stated otherwise, all materials described in this work were prepared on purpose-built experimental setups described in the thesis. Commercial colloidal quantum dots from Sigma-Aldrich and samples from the University of Toronto (Canada) served as certified reference samples.

Microscopy studies of materials and structures used an NTEGRA Thermo scanning probe microscope (NT-MDT), a MIRA3 scanning electron microscope (TESCAN), and a JEM-2100F transmission electron microscope (JEOL). Transmission spectra were measured on PE-5400UV (Ekros) and SF-256BIK (LOMO Photonica) spectrophotometers. Photoluminescence spectra were studied with a setup based on an MDR-206 monochromator (LOMO Photonica), a silicon photodiode with a preamplifier, and a 405 nm solid-state laser. The radiation hardness experiments used a gamma irradiator with a ^{137}Cs radionuclide source, a photon energy of 661 keV, and a dose rate of 1.7 Gy/s.

Presentation of the results. The results of the thesis were presented and discussed at international scientific and technical conferences (Nanostructures: Physics and Technology (2013), IEEE NW Russia Young Researchers in Electrical and Electronic Engineering Conference (EIConRusNW) (2015, 2016), Saint-Petersburg OPEN (2015, 2016), Photonics North (2015, 2016), Photonic Colloidal Nanostructures: Synthesis, Properties, and Applications (2016), E-MRS Fall Meeting (2017)) and at all-Russian conferences, seminars, and schools (Current Problems of Nano- and Microelectronics (Ufa, 2012), Diagnostics of Nanomaterials and Nanostructures (Ryazan, 2012–2015), Scien-

tific Youth School on Solid-State Electronics (2009–2015), All-Russian Youth Conference on the Physics of Semiconductors and Nanostructures, Semiconductor Opto- and Nanoelectronics (2015), Russian Youth Conference PhysicA.SPb (2012–2015), All-Russian Inter-University Scientific and Technical Conference of Students and Postgraduates “Microelectronics and Informatics” (Moscow, 2013), Nanostructured Materials and Conversion Devices for Solar Energy (2014), 3rd All-Russian Conference on Photonics and Information Optics (Moscow, 2014), Congress of Young Scientists (2014), Interdisciplinary Youth Scientific Forum “New Materials. Days of Science” (2015)).

Author’s personal contribution. The author developed automated experimental setups and characterization equipment, and proposed sets of process steps for the following tasks: colloidal synthesis of nanoparticles by hot injection, synthesis of nanoparticles in a flow of immiscible liquids, layer-by-layer deposition of film nanostructures, measurement of static light-scattering indicatrices of colloidal solutions and thin films, and measurement of the optical characteristics of layered structures. The author synthesized the nanoparticles, deposited the layers, fabricated the structures, studied the materials by absorption and luminescence spectroscopy and static light scattering, and analyzed and systematized the results.

Publications. 25 works on the topic of the thesis have been published: 10 papers in leading peer-reviewed journals listed by the Higher Attestation Commission (VAK) of Russia, 20 papers in journals indexed in Web of Science and Scopus, three Russian utility model patents, 1 textbook, and 1 monograph.

Structure and scope of the thesis. The Russian original of the thesis comprises 131 pages. It consists of an introduction, four chapters with conclusions, a conclusion, and a list of references with 139 items. It contains 79 figures and 7 tables.

Main Content of the Work

The **introduction** justifies the relevance of the work, states the aim and objectives of the research, shows the practical significance of the results, and formulates the main statements for defense.

The **first chapter** reviews the literature on the basic physical and chemical properties of metal and semiconductor nanoparticles of different chemical composition, which determine the physical and technological principles of the colloidal synthesis of monodisperse quantum dots and plasmonic nanoparticles. Scaling up colloidal synthesis is considered the key problem in making nanoparticle synthesis more efficient.

The **second chapter** deals with the practical implementation of the physical and technological principles of the synthesis of semiconductor and metal nanoparticles described in the first chapter, and with the monitoring of the synthesis and characterization of its products.

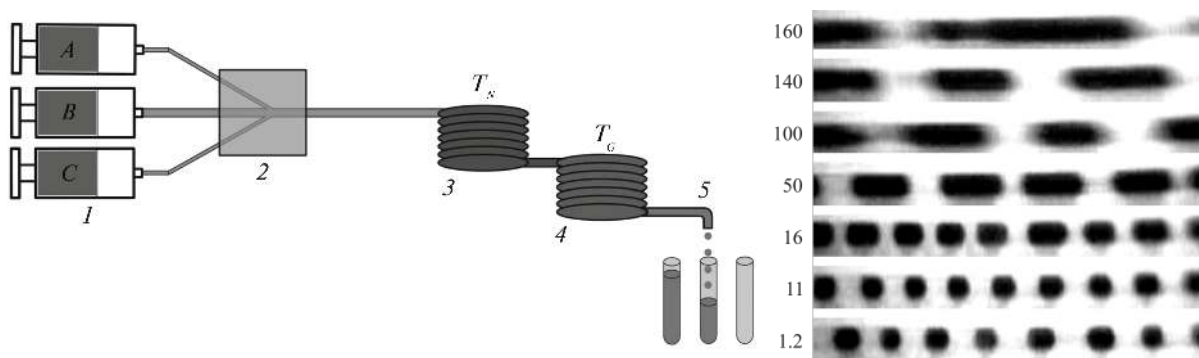


Figure 1. Design of the flow reactor (left) and evolution of the segmented flow as the flow rates ($\mu\text{L/s}$) of the precursors and the carrier liquid are changed simultaneously

The problem of a method for the mass synthesis of monodisperse nanoparticles is solved by moving from synthesis in a flask (the hot-injection method) to flow synthesis. In the hot-injection method, sources of ions A and B are rapidly mixed at the nucleation temperature T_N (for metal particles, the role of B is played by a reducing agent). Because the AB monomers quickly reach supersaturation throughout the volume, a large number of nuclei form simultaneously, and the material then condenses on them at the maintained growth temperature T_G . The particle surface is stabilized by surfactants, which create a barrier to aggregation and passivate surface states.

In flow synthesis, the reaction medium is a liquid flow passing through zones at the temperatures T_N and T_G : the nucleation and growth zones. The corresponding times are set by the flow rates of the liquids from the syringe pumps containing components A and B , and by the path length of the particles in each zone. A continuous flow in a thin channel, especially of viscous liquids, develops a parabolic velocity profile, which widens the particle size distribution; in addition, it gradually clogs the channel. These problems are solved by segmenting the flow with a carrier liquid C that is immiscible with the reaction medium and wets the channel walls. As a result, a reactor design was proposed (Fig. 1) that consists of syringe pumps 1, a mixer 2, temperature-controlled zones 3 and 4, and a polytetrafluoroethylene capillary 5 with an inner diameter of 0.9 mm (the diameter sets the maximum droplet size). The engineering solution is protected by Russian patent 166323.

Fig. 1 shows the transition from continuous flow at precursor and carrier flow rates above 150 $\mu\text{L/s}$ to a segmented liquid-liquid flow. On the one hand, lower flow rates give more uniform droplets. On the other hand, at flow rates of ~ 1 $\mu\text{L/s}$, mixing slows down. Flow rates of 10 $\mu\text{L/s}$ were chosen as working rates, as they give droplets of equal volume and uniform mixing in each subsystem.

In electronic devices, semiconductor materials are usually in the solid state, so an important part of the research was the development of a deposition method for the synthesized nanocrystals. In addition to spin coating, traditionally used for this purpose, we proposed an automated setup and the

corresponding software for successive ionic layer adsorption and reaction and electrochemical atomic layer deposition under a variable external electric field (Russian patent 156478). This solution made it possible to deposit not only nanoparticles, as demonstrated for CsPbBr₃ nanocrystals, but also layers from films.

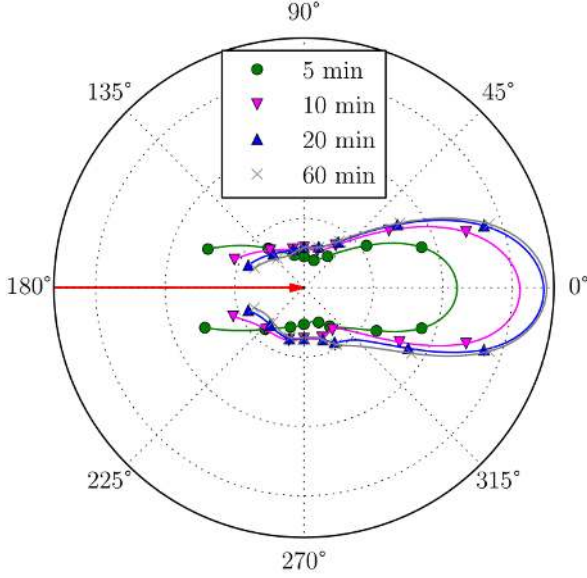


Figure 2. Experimental light-scattering indicatrices of SiO₂ particles recorded at different times after nucleation

For example, Fig. 2 shows a set of indicatrices for the growth of SiO₂ nanoparticles by the Stöber method [3]. It shows the transition from the shape of Rayleigh scattering at the early stages of growth to the shape described by the Mie model in later indicatrices. The end of nanoparticle growth is identified by the moment when the shape and size of the indicatrices stop changing.

Thus, the physical and technological solutions described in the second chapter improve the monitoring of the synthesis of semiconductor and metal nanoparticles and allow the colloidal synthesis of monodisperse nanoparticles to be scaled up, with subsequent ordering of the particles on substrates. However, as follows from the literature review in the first chapter, the efficiency of the final devices can also be increased by searching for new, previously unstudied materials, in particular ternary compounds.

Therefore, the **third chapter** considers the properties of a new class of nanomaterials: perovskite QDs of the ternary cesium lead halides CsPbX₃ (X = Cl, Br, I) and their solid solutions. They reach a high quantum efficiency without inorganic shells of wide-gap semiconductors, which are usually needed for an acceptable quantum efficiency of A^{II}B^{VI} and A^{IV}B^{VI} colloidal QDs.

The upper part of Fig. 3 shows the perovskite crystal lattice with the general formula ABX₃. In CsPbX₃, the A site is occupied by the Cs⁺ cation, B by the lead cation Pb²⁺, and X by a halide anion: chloride, bromide, or iodide. One of the criteria used to assess the stability of the structure is

Preparing colloidal solutions and depositing nanoparticle films from them requires continuous monitoring of nucleation/growth and deposition. For this, we proposed and built an experimental setup for studying the synthesis of nanostructured materials by the method of light-scattering indicatrices, i.e., the angular distributions of the intensity of electromagnetic radiation scattered by particles. The method was shown to allow real-time monitoring of the synthesis of various types of nanoparticles and layers (the engineering solution is protected by Russian patent 167044).

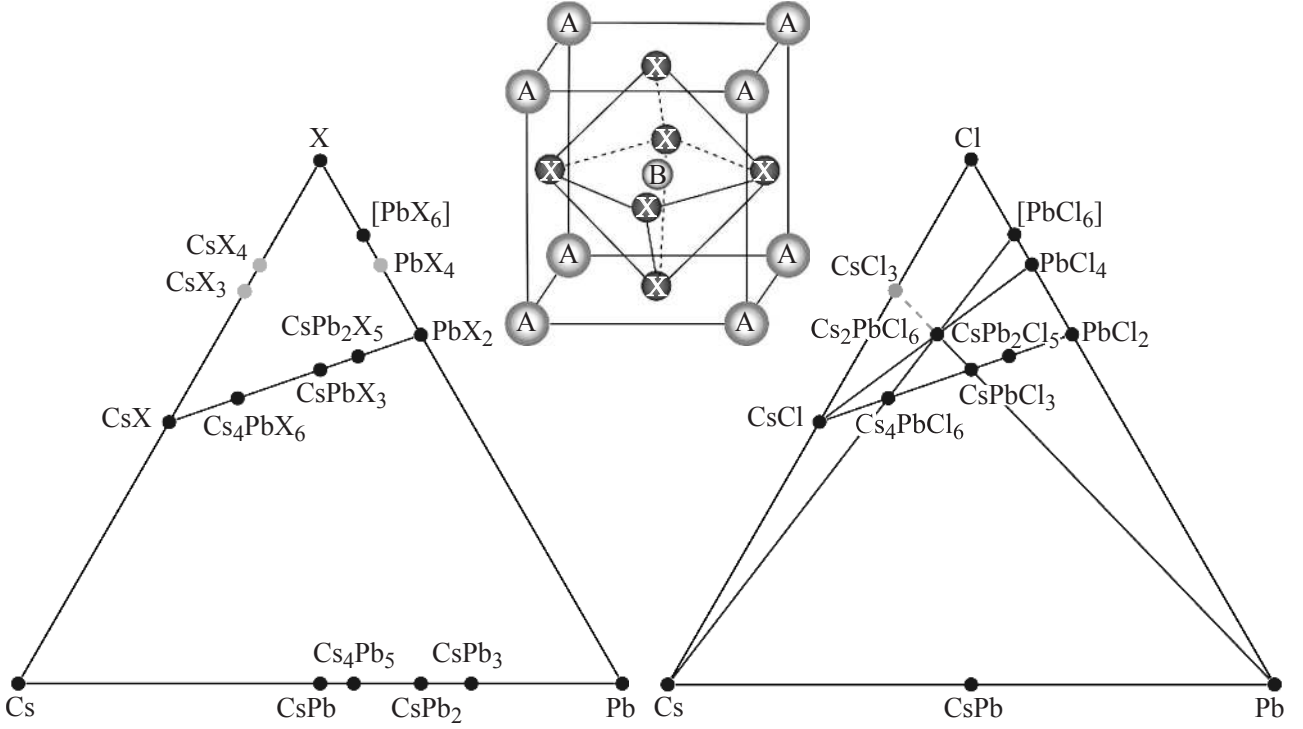


Figure 3. Perovskite crystal lattice, main compounds of the ternary Cs–Pb–X systems on the Gibbs triangle, and partial triangulation of the system, using CsPbCl₃ as an example

the Goldschmidt tolerance factor $t = (r_A + r_X)/(\sqrt{2}(r_B + r_X))$, where r_A , r_B , r_X are the ionic radii. The lattice geometry implies that $t = 1$ in an ideal cubic crystal. In the range $t = 0.7$ – 0.9 , to which cesium lead halides belong, other crystal symmetries are possible: besides the cubic perovskite, tetragonal, orthorhombic, and monoclinic crystal systems are also found.

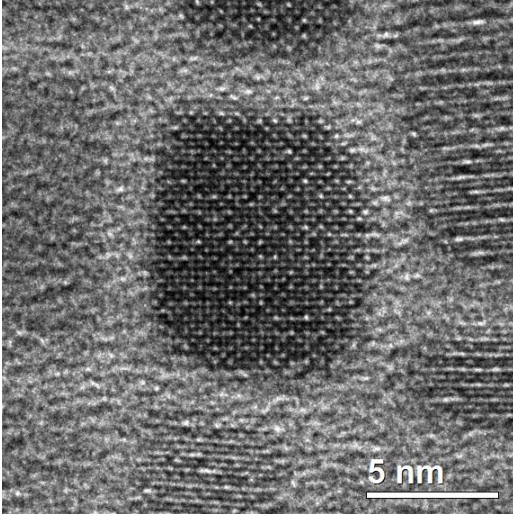


Figure 4. TEM image of a single CsPbBr₃ nanocrystal

For low-dimensional forms of the material, however, perovskite nanocrystals of the stable cubic crystal system with the $Pm\bar{3}m$ structure can be observed, as confirmed by X-ray diffraction and high-resolution transmission electron microscopy (TEM). Fig. 4 shows both a strictly periodic crystal lattice and organic layers between individual nanocrystals, whose thickness corresponds to the length of the organic ligand molecules capping the nanocrystals: oleic acid C₁₇H₃₃COOH and oleylamine C₁₈H₃₅NH₂.

These nanocrystals were first obtained by colloidal synthesis in 2015 [4]. The ternary Cs–Pb–X systems are still insufficiently studied from the point of view of physicochemical analysis. In this work, the ternary Cs–Pb–X systems (X = Cl, Br, I) were analyzed physicochemically by triangulation for the first time. Fig. 3 illustrates the result of the analysis on the Gibbs triangle. The analysis showed that for none of the halogens

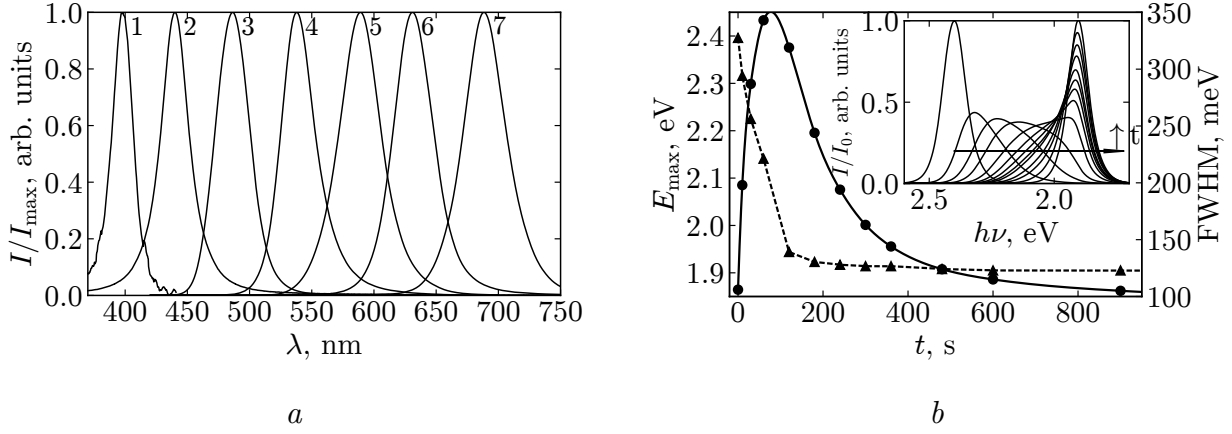


Figure 5. a) PL spectra of CsPbX₃ nanocrystals with an edge length of ~ 10 nm: 1, CsPbCl₃; 2, CsPb(Cl_{1-x}Br_x)₃, $x = 0.5$; 3, CsPbBr₃; 4–6, CsPb(Br_{1-x}I_x)₃, $x = 0.33, 0.5$, and 0.67 , respectively; 7, CsPbI₃; b) dynamics of the PL spectra during Br–I anion exchange: time dependence of the photon energy of the PL maximum ($\blacktriangle$) and of the PL line FWHM ($\bullet$). Inset: evolution of the PL spectrum with time

taken separately does the triangulation contain the sections CsX₃–PbX₄ or CsX₄–PbX₄, and the main section, present in all the systems, is CsX–PbX₂. This section contains the composition point of the CsPbX₃ perovskites, as well as the points of the thermodynamically less stable compounds Cs₄PbX₆ and CsPb₂X₅. An important element of the triangulation was found to be the [PbX₆] composition point, which corresponds to an octahedron of a central lead ion Pb²⁺ coordinated by six halogen atoms (Fig. 3). This element, unusual for classical triangulation, plays a decisive role in the growth of nanocrystals from solution. The ratio of the halide ion concentrations determines the photoluminescence of the material (Fig. 5a), so that the PL maximum can be tuned across the entire visible range.

One of the features of halide perovskites is fast anion exchange: the formation of solid solutions through the chemical reaction of a cesium lead halide with a source of another halogen. For these perovskites, the process occurs under ambient conditions. This is of great practical importance. Nanocrystals of a certain chemical composition can be synthesized at elevated temperature, and then the PL maximum can be moved to a desired position by adding a source of another halogen at room temperature, without further heating.

To study the dynamics of anion exchange, a concentrated colloidal solution of CsPbBr₃ nanoparticles, purified from reaction products beforehand, was mixed with a comparable amount of PbI₂ precursor. After mixing, the PL spectrum of the solution was measured continuously. Fig. 5b shows the dynamics of the PL spectrum, i.e., the energy of the PL maximum and the line full width at half maximum (FWHM). Both dependences split into two segments with a break point at the same time, 120 s. The shape of these dependences suggests that the key change results from processes at the surface of the perovskite nanocrystals. The smooth shift of the PL spectrum and the

absence of distinct features during the shift indicate sharing (hybridization) of the corresponding electronic states. The type of chemical compound used as the halogen source was shown to be critical: adding I_2 instead of PbI_2 in the same experiment only etches the nanocrystal surface and gradually quenches the PL.

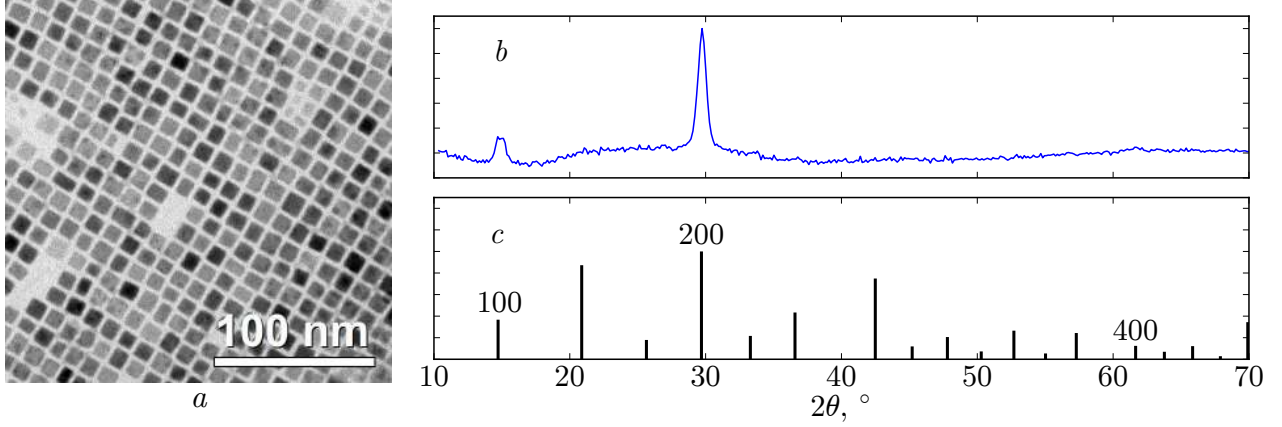


Figure 6. a) TEM image of an ordered ensemble of $CsPbBr_3$ nanocrystals, b) experimental diffraction pattern of an ordered structure deposited on glass, c) calculated stick pattern of $CsPbBr_3$ $Pm\bar{3}m$ with the reflections observed in the experimental pattern highlighted

The monodispersity of cesium lead halide nanoparticles, their cubic shape, and the weak binding of organic ligands to the nanocrystal surface make oriented attachment possible. This is confirmed by TEM data (Fig. 6a) for a layer of $CsPbBr_3$ nanocrystals on a graphene-coated TEM grid, and by XRD data for a layer of the same nanocrystals deposited layer by layer on a glass substrate (Fig. 6b, layer thickness ~ 100 nm). Comparison of the experimental diffraction pattern with the calculated values shown in Fig. 6c shows that the layers remain strongly (100)-oriented as they grow thicker.

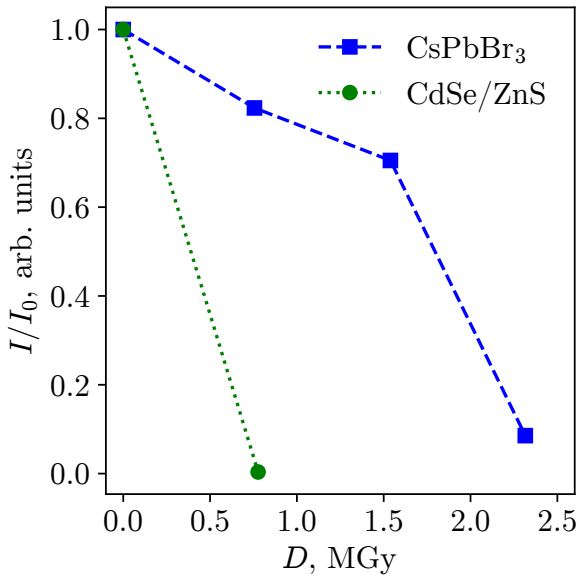


Figure 7. PL intensity of colloidal QDs of different composition versus γ -radiation dose

The physicochemical features of the cesium lead halide system discussed above, and the absence of defect-related bands in the PL spectra, suggest that the system may be resistant to defect formation. The fast anion exchange indicates that the system tends to restore the anionic octahedra, the main structural units responsible for PL. To test this, we studied the effect of γ -radiation from a ^{137}Cs source on the PL of cesium lead halide perovskite nanocrystals. Irradiation took place in a chamber inside a shielded enclosure, at an absorbed dose rate (in water) of

1.7 Gy/s. The samples were placed in the chamber in glass vials and irradiated for set times, accumulating an absorbed dose (in water) of up to 2.3 MGy in total. About a day passed between exposures, during which the samples were prepared and their spectra measured. The experiment showed that the PL of CsPbBr_3 is more stable than that of commercial CdSe/ZnS nanocrystals (Fig. 7). These results show that CsPbBr_3 nanocrystals are promising as a luminescent material in high-radiation environments, including, with a suitable protective shell, as biomarkers for anticancer radiotherapy.

Despite these advantages, however, cesium lead halides also have drawbacks associated with their ionic bonding, which makes these materials sensitive to moisture.

The **fourth chapter** addresses the increase in the luminescence intensity of metal chalcogenide quantum dots through the interaction of QDs with plasmonic nanoparticles. To study plasmonic enhancement, Ag/SiO_2 core/shell particles were synthesized by colloidal chemistry. Monodisperse particle cores were obtained by colloidal synthesis with two reducing agents of different nature: ascorbic acid and sodium citrate. Ag particles synthesized by this method, without a dielectric shell, show an LSPR at 405 nm. The shells were grown by the modified Stöber method [3], with the Ag nanoparticles from the previous step as cores. Different dielectric shell thicknesses h_s were obtained by adding different

amounts of tetraethyl orthosilicate to the colloidal solution of Ag particles; it turns into SiO_2 through hydrolysis and polycondensation. The particle diameter, determined from the plasmon resonance position and confirmed by scanning electron microscopy, was $d = 17$ nm. As h_s increased, the plasmon resonance maximum shifted slightly to lower frequencies because the permit-

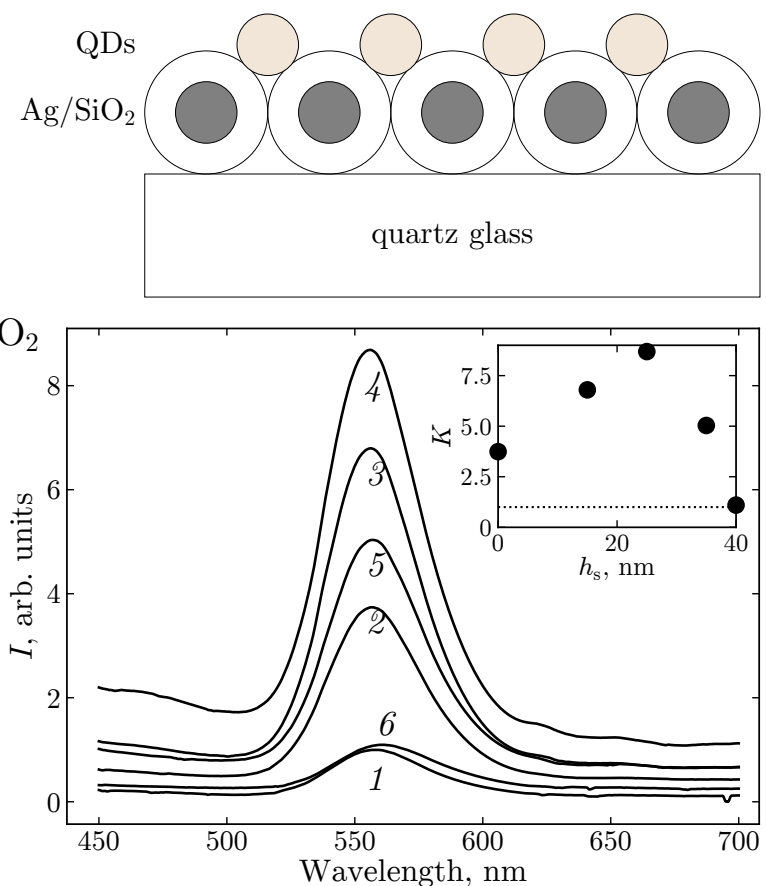


Figure 8. Schematic of the structures for studying plasmonic enhancement, and PL spectra of CdSe/ZnS QDs without (1) and near plasmonic particles with a shell thickness $h_s = 0$ (no shell) (2), 15 (3), 25 (4), 35 (5), and 40 nm (6). Inset: enhancement factor versus shell thickness

tivity of the surrounding medium changed.

The Ag/SiO₂ particles were deposited on a quartz substrate, and a QD layer was then deposited on top of this layer from a colloidal solution, giving the structure shown schematically in Fig. 8. The QD layers were deposited at a ratio of the surface concentrations of QDs and Ag/SiO₂ nanoparticles at which the interaction between QDs can be neglected and mainly the interaction of the semiconductor particles with the underlying layer of Ag/SiO₂ nanoparticles is observed. All types of QDs were much smaller than the plasmonic particles, so a single QD most likely interacts with a single plasmonic nanoparticle. The PL of the sample was excited from the QD layer side with a solid-state laser whose wavelength matches the plasmon resonance of the particles.

To describe the interaction of the particles, we introduced the PL enhancement factor $K = \max\{I(\lambda)\}/\max\{I_0(\lambda)\}$, where $I(\lambda)$ and $I_0(\lambda)$ are the PL spectra of the QD layer with and without the plasmonic nanoparticle layer, respectively. The PL spectra of the structures with CdSe/ZnS QDs, normalized to the PL maximum of the structure without plasmonic particles, are shown in Fig. 8. The spectral shape and peak position do not change substantially in the presence of plasmonic nanoparticles, while the intensity first increases considerably with the shell thickness and then decreases. The inset in Fig. 8 shows the enhancement factor versus the shell thickness: the largest enhancement, $K = 8.7$, is observed at a shell thickness of $h_s = 25$ nm. Note that the PL intensity grows together with the background scattering caused by the larger size of the Ag/SiO₂ plasmonic nanoparticles. This scattering is apparently an additional reason for the PL increase. Note also that at $h_s = 40$ nm the PL intensity falls to the initial level. Thus, the PL enhancement is maximal at an optimal interparticle distance. The rapid decrease of the enhancement factor with increasing shell thickness reflects the near-field nature of PL enhancement by metal nanoparticles.

The **conclusion** presents the main scientific and practical results of the work and gives references to examples of the use of the obtained nanostructures as biomarkers, as components of organic–inorganic light-emitting field-effect transistors, and in quantum dot layers showing optical orientation and alignment of excitons.

Main Results of the Work

1. Within a single technological approach, a technology was developed and implemented for the synthesis of monodisperse nanocrystals: colloidal quantum dots of metal chalcogenides and cesium lead halides, and silver-based plasmonic nanoparticles.
2. Automated experimental setups were proposed and built for 1) nanoparticle synthesis in a flow of immiscible liquids, 2) deposition of nanoparticle layers from colloidal solutions, and 3) optical monitoring of nanoparticle

synthesis and nanoparticle layer deposition by measuring light-scattering indicatrices. These engineering solutions are protected by three Russian utility model patents.

3. Scientifically grounded synthesis conditions were developed for nanocrystals capable of oriented attachment, of CsPbX_3 ($X = \text{Cl}, \text{Br}, \text{I}$) perovskites and solid solutions based on them. They rely on the concept of the decisive role of the $[\text{PbX}_6]^{4-}$ anionic octahedral complex.
4. Ag nanoparticles 20 nm in diameter with a 25 nm SiO_2 shell increased the photoluminescence intensity of a CdSe/ZnS layer more than eightfold compared with the same layer without plasmonic nanoparticles.
5. The synthesized luminescent CsPbBr_3 nanoparticles were shown to be more radiation-hard than commercial CdSe/ZnS nanoparticles.
6. XRD demonstrated that layers of halide perovskite nanocrystals up to several hundred nanometers thick can be deposited with a strong (100) orientation.

Main Publications on the Topic of the Thesis

1. Ryzhov O.A., **Matyushkin L.B.**, Moshnikov V.A., Aleksandrova O.A. Reactor for the synthesis of nanostructures. RU Patent 166323; filed 24.12.2015; published 20.11.2016.
2. Moshnikov V.A., Maksimov A.I., Aleksandrova O.A., **Matyushkin L.B.**, Ryabko A.A. Device for studying materials by the light-scattering indicatrix method. RU Patent 167044; filed 14.06.2016; published 20.12.2016.
3. **Matyushkin L.B.**, Khondryukov D.V., Aleksandrova O.A. Automated setup for depositing thin-film coatings of nanoscale thickness by molecular layering. RU Patent 156478; filed 21.04.2015; published 10.11.2015.
4. Aleshin A.N., Shcherbakov I.P., Gushchina E.V., **Matyushkin L.B.**, Moshnikov V.A. Solution-processed field-effect transistors based on polyfluorene–cesium lead halide nanocrystals composite films with small hysteresis of output and transfer characteristics // *Organic Electronics*. 2017. V.50. P.213–219.
5. Nestoklon M.O., Goupalov S.V., Dzhioev R.I., Ken O.S., Korenev V.L., Kusrayev Yu.G., Sapega V.F., de Weerd C., Gomez L., Gregorkiewicz T., Lin J., Suenaga K., Fujiwara Y., **Matyushkin L.B.**, Yassievich I.N. Optical orientation and alignment of excitons in ensembles of inorganic perovskite nanocrystals // *Physical Review B*. 2018. V.97, No. 23. P.235304.
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